The Classical Agent

Volatility targeting. Stateless. The emotion-free control.

1. Decision rule

Trailing realised volatility, annualised, over the n returns *strictly before* day t:

$$\hat{\sigma}_t = \sqrt{A} \cdot \mathrm{sd}\big(r_{t-n}, \ldots, r_{t-1}\big) \tag{1}$$

Exposure scales inversely, targeting $\sigma^\star$:

$$w_t = \mathrm{clip}\left(\frac{\sigma^\star}{\max(\hat{\sigma}_t, \varepsilon)},\ w_{\min},\ w_{\max}\right) \tag{2}$$

$$V_t = V_{t-1}(1 + w_t r_t) \tag{3}$$

$$\mathrm{DD}_t = \big(\max_{u \le t} V_u - V_t\big) \big/ \max_{u \le t} V_u \tag{4}$$

where $\mathrm{clip}(x, a, b) = \min\{\max\{x, a\}, b\}$.

2. Variables

Symbol	Meaning	Value
t	Trading-day index	—
r_t	Total return on day t (dividend-inclusive)	data
$\hat{\sigma}_t$	Estimated annualised volatility, data through $t-1$	computed
w_t	Exposure for day t	computed
V_t	Portfolio value after day t	$V_0 = 100$
DD_t	Drawdown from running peak; reporting only	computed
n	Volatility lookback (trading days)	20
$\sigma^\star$	Annualised volatility target	0.10
A	Annualisation factor	252
ε	Numerical floor on $\hat{\sigma}_t$	10^{-4}
$w_{\min}, w_{\max}$	Exposure bounds	0.05, 1.00

3. Properties

Stateless. $w_t = f(r_{t-n}, \ldots, r_{t-1})$. Nothing carries forward; a return leaving the window ceases to influence the agent.

Symmetric. Returns enter (1) through squared deviations, so r and $-r$ are identical inputs.

No interior. $\hat{\sigma}_t$ estimates a property of the world, not a state of the agent. Hence no welfare counterpart.

No look-ahead. The window terminates at $t-1$.